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کابل پوهنتون



Microsoft Excel

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Introduction to Excel Functions

- ❖ Microsoft Excel functions are predefined formulas that perform specific calculations by using values you provide as inputs.
- ❖ They are designed to simplify complex tasks, saving you the time of building a manual formula step-by-step.
- ❖ Common categories: Math, Statistical, Logical, Text, Date & Time, Lookup & Reference.
- ❖ Real-world uses: grading systems, financial analysis, inventory management, attendance tracking.

What is a Formula?

- A formula is an expression used to calculate a value in a cell.
- Rules:
- Must begin with '='
- Can include numbers, cell references, operators (+ - * /), and functions
- Examples: =A1+A2 (addition) =A1*B1 (multiplication) =(A1+B1)/2 (average without function)

What is a Function?

- A function is a predefined formula designed for a specific task.

Advantages:

- Saves time
- Reduces complexity
- Minimizes errors Example: =SUM(A1:A10) replaces manual addition of each cell

Structure of a Function

- General format: =FUNCTION_NAME(argument1, argument2,...) Arguments can be:
- Numbers (10, 20)
- Cell references (A1, B2)
- Ranges (A1:A10)
- Tip: Use commas to separate arguments.

AVERAGE Function (Concept)

- AVERAGE calculates the arithmetic mean.
- Formula concept: Sum of values $\div$ number of values.

Used in:

- Student grading
- Sales analysis
- Performance evaluation

AVERAGE Syntax

=AVERAGE(number1, number2,...) Examples: =AVERAGE(A1:A10)
=AVERAGE(A1,A2,A3)

Note: Ignores empty cells and text.

AVERAGE Detailed Example:

- Data: 10, 20, 30, 40 Calculation: $(10+20+30+40)/4 = 25$ Excel formula: =AVERAGE(A1:A4) Important:
- If a cell contains text, it is ignored
- If zero exists, it is included

COUNT Function

- COUNT counts cells containing numeric values only.

It does NOT count:

- Text
- Empty cells
- Logical values (TRUE/FALSE)

COUNT Syntax:

=COUNT(value1, value2,...) Examples: =COUNT(A1:A10) =COUNT(A1, A2, A3)

COUNT Detailed Example

- Data: 5, "Ali", 10, "", TRUE Result: = 2 Explanation: Only 5 and 10 are numeric
Tip: Use COUNTA to count all non-empty cells

MAX Function

- MAX finds the largest value in a dataset. Used for:
- Highest marks
- Maximum profit
- High temperature

MAX Syntax:

=MAX(A1:A10) Can also use: =MAX(10,20,30)

MAX Detailed Example:

Data: 5, 20, 15, 50 Result: 50 Use case: Finding top-performing student or best sales day

MIN Function

- MIN finds the smallest value. Used for:
- Lowest marks
- Minimum cost
- Least temperature

MIN Syntax:

=MIN(A1:A10)

Example:

Data: 5, 20, 15, 2 Result: 2 Use case: Identifying weakest performance or lowest value

SUM Function

- SUM adds values quickly. Examples: =SUM(A1:A10) =SUM(A1,A2,A3)
Advantages:
- Faster than manual addition
- Works with large datasets

Logical Functions

- Which are used to check logic(Condition).
- Logical functions are used to perform decision-making in Excel. They evaluate conditions and return results based on TRUE or FALSE.
- These functions are very important in real-life scenarios such as grading, filtering data, and automation.

IF Function

- IF performs logical decision-making.
- IF is the most commonly used logical function.
- Syntax: =IF(condition, value_if_true, value_if_false) It checks whether a condition is true or false.
- Example: =IF(A1>50,"Pass","Fail")

Nested IF Function

- Nested IF means using multiple IF functions inside one formula.

Example: `=IF(A1>=80,"A", IF(A1>=60,"B", IF(A1>=40,"C","Fail")))`

Use case: Grading system with multiple levels.

AND Function

AND checks multiple conditions and returns TRUE only if ALL conditions are true.

Syntax: =AND(condition1, condition2,...)

Example: =AND(A1>50, B1>50) If both conditions are true → TRUE Otherwise → FALSE

Example

Data:

- Fever: C°39
- Pulse:110
- Blood Pressure:140

Information:

The Patient has Fever, he might be ill.

Excel Example: =IF(AND(Fever>=39,Pulse>=110,Blood Pressure>=140),"Ill",
"Normal")

OR Function

- OR checks multiple conditions and returns TRUE if ANY condition is true.
Syntax: =OR(condition1, condition2,...)
- Example: =OR(A1>50, B1>50) If one is true → TRUE

Adding Comments

Steps:

1. Select a cell
2. Go to Review tab
3. Click New Comment

Type message Use cases:

1. Teacher feedback
2. Team discussion

Track Changes

Track Changes records all edits. Shows:

- Who made changes
- What was changed
- When it was change

Enabling Track Changes

- Steps:
- Review tab
- Track Changes
- Highlight changes Important: Useful in teamwork and document approval

Practical Exercise

Students should:

- Create dataset of marks
- Apply:
 - AVERAGE
 - COUNT
 - MAX
 - MIN
- Use IF for grading
- Add comments
- Track changes